



Fault-Propagate Pattern Based DFA on PRESENT and PRINTcipher

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Abstract: This article proposes an enhanced differential fault analysis (DFA) method named as fault-propagation pattern-based DFA (FPP-DFA). The main idea of FPP-DFA is using the FPP of the ciphertext difference to predict the fault location and the fault-propagation path. It shows that FPP-DFA is very effective on SPN structure block ciphers using bitwise permutation, which is applied to two block ciphers. The first is PRESENT with the substitution-permutation sequence. With the fault model of injecting one nibble fault into the r -2nd round, on average 8 and 16 faults can reduce the key search space of PRESENT-80/128 to $2^{14.7}$ and $2^{21.1}$, respectively. The second is PRINTcipher with the permutation-substitution sequence. For the first time, it shows that although the permutation of PRINTcipher is secret key dependent, FPP-DFA still works well on it. With the fault model of injecting one nibble fault into the r -2nd round, 12 and 24 effective faults can reduce the key search space of PRINTcipher-48/96 to $2^{13.7}$ and $2^{22.8}$, respectively.

Key words: fault-propagation pattern; differential fault analysis; bitwise permutation; SPN block cipher; PRESENT; PRINTcipher

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0 Introduction

PRESENT^[1] and PRINTcipherX^[2] are two ultra-lightweight block ciphers presented in CHES 2007 and CHES 2010, respectively. Both ciphers apply the SPN structure and bitwise permutation, except that the substitution-permutation (S-P) sequence is different and PRINTcipher adopts a secret key-dependent permutation. The designers analyzed the security of them with respect to the main known cryptanalytic methods, which showed that they were secure. However, the security of the ciphers against fault attacks was not considered. Fault attacks were firstly introduced by Boneh *et al.*^[3] on RSA public key cryptosystem in 1997. Shortly after, Biham and Shamir^[4] proposed an attack on the secret key cryptosystems, called differential fault analysis (DFA), which combined the ideas of differential cryptanalysis with fault attacks. Since then, various cryptosystems have been attacked by DFA, such as ECC^[5], AES^[6,7], Camellia^[8,9], MIBS^[10], and RC4^[11]. However, to the best of our knowledge, few works, except Refs. [12] and [13], have been done on DFA against PRESENT, and there is no publication about DFA on PRINTcipher.

Li and Gu^[12] proposed the first DFA on PRESENT-80. They adopted the fault model of injecting one nibble (4-bit) fault between the $r-2$ nd round and the $r-1$ st round substitution input. They proved that on average 40 to 50 effective faults could recover the 64-bit post-whitening key. Wang *et al.*^[13] proposed the first DFA on the key schedule of PRESENT-80. Suppose one nibble fault was injected into the 80-bit intermediate updated key register while generating the last round key and the post-whitening key; they used about 64 pairs of correct/faulty

samples to obtain the 51-bit key.

In this article, we present an enhanced DFA method named as fault-propagation pattern-based DFA (FPP-DFA). In DFA on block ciphers with SPN structure, how to recover the S-box input difference from the S-box output difference is the critical issue. It shows that FPP-DFA can predict the fault-propagation path to recovery the S-box input difference and enhance the traditional DFA. In particular, we conclude that FPP-DFA is quite efficient on SPN block ciphers with bitwise permutation, which is applied to PRESENT and PRINTcipher.

The rest of the article is structured as follows: Section 1 describes the main idea of the proposed FPP-DFA. Sections 2 and 3 present the detailed information of FPP-DFA attacks on PRESENT and PRINTcipher, respectively. Section 4 concludes the article.

1 FPP-DFA

1.1 Traditional DFA

In fault attacks, the adversary usually injects a fault in one operation of the last few rounds, and then several bits of the ciphertexts become faulty.

As shown in Fig. 1, the adversary injects one byte fault into the third byte of the 10th AES round substitution layer input X_2 , X_2 becomes faulty X_2^* . After the substitution layer Y_2 , the correct S-box lookup result of X_2 becomes faulty Y_2^* . Suppose f_1 and f_2 denote the S-box input and output fault difference of X_2 . Let a be the correct S-box index, $a=X_2$, a^* be the faulty S-box index, $a^*=X_2^*$, then $f_1=a \oplus a^*$. Let $S[]$ denote the S-box lookup function, then f_2 can be described as

$$S[a] \oplus S[a \oplus f_1] = f_2 \quad (1)$$

The target of DFA is to recover a or $S[a]$. Usually, $S[a]$ has close relationship with the round key and the ciphertexts. The round key can be retrieved by $S[a]$ and ciphertexts easily. After that, the adversary can inject faults in other locations or rounds to analyze more keys until the master key can be retrieved.

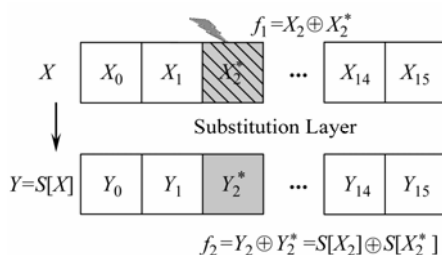


Fig. 1 Fault in the 10th round of AES

As to SPN structure block cipher, f_1 is usually unknown. How to predict f_1 is the key point of DFA.

1.2 Proposed FPP-DFA

The main idea of FPP-DFA is to use the FPP of the ciphertext difference to infer the fault location and predict the fault-propagation path. As to SPN block ciphers with bitwise permutation, the recovery of the fault-propagation path means the recovery of the S-box input difference, which is the most challenging part of DFA on SPN block ciphers. Based on this, the traditional DFA technique can be used for further key analysis. The procedure of FPP-DFA is demonstrated in Fig. 2. Next, we present the main phases of the FPP-DFA.

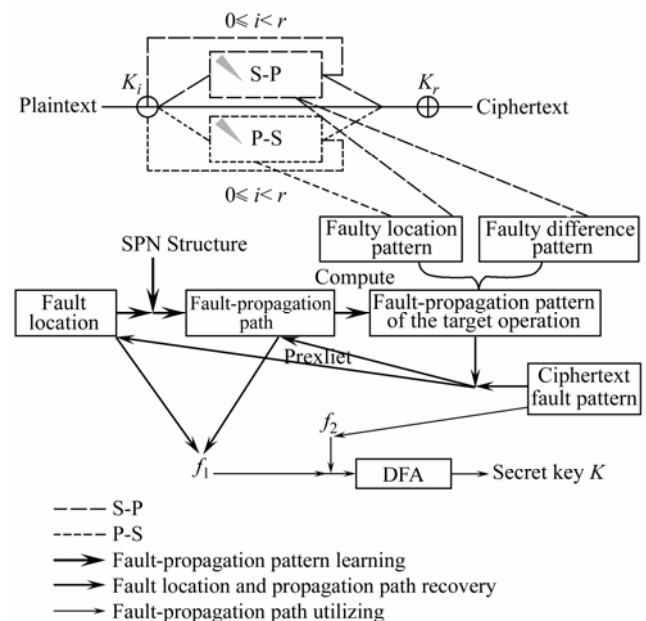


Fig. 2 Procedure of FPP-DFA

S-P: substitution-permutation; P-S: permutation-substitution

① FPP learning

The adversary learns the FPP of the targeted operation and then builds the distinguisher for the enumerable faulty locations based on the fault model and the feature of the substitution and permutation operations. As to FPP-DFA on SPN block ciphers with S-P sequence, the targeted operation is the last round substitution layer output. Two kinds of FPP are applied. The first is the faulty-location (index) pattern, and the second is the fault-difference pattern. As to FPP-DFA on SPN block cipher with P-S sequence, the targeted operation is the last round permutation layer output and only the faulty location pattern is considered.

② Fault location and propagation path recovery

To recover the fault-propagation path, the adversary first computes the real FPP of the targeted operation

from the fault pattern of the ciphertext difference. Then, according to the FPP distinguisher for the enumerable fault locations, the very or several fault locations and related fault-propagation paths can be predicted. As to current SPN structure block ciphers with bitwise permutation layer, for a single byte (nibble) fault injection, if the fault does not propagate to the same byte (nibble) twice, the recovery of the fault-propagation path means that the S-box input difference f_1 of the propagated rounds can be obtained.

③ Fault-propagation path utilizing

After the above two phases, the fault-propagation path can be predicted and used for further key recovery. For example, the S-box input difference f_1 of the final target substitution layer can be recovered. As the S-box output difference f_2 can be computed easily by the ciphertext difference, combining the Eq. (1) and the traditional DFA technique, the secret key can be recovered.

Next, we take PRESENT and PRINTcipher as two examples and apply the FPP-DFA to prove its feasibility and efficiency.

2 FPP-DFA on PRESENT

2.1 PRESENT Algorithm

PRESENT^[4] is a 31-round SPN structure block cipher with block size of 64 bits, and the cipher is described in Fig. 3. It supports 80- and 128-bit secret key. Firstly, the plaintext is the input of the first round. After 31-round iterations, the 31st-round output Xored with the post-whitening key K^{32} is the ciphertext.

Each round is composed of the following three steps.

① addRoundKey (AK): At the beginning of each round, 64-bit output of the last round is Xored with the subkey.

② sBoxlayer (SL): 16 identical 4-bit to 4-bit S-boxes are used in parallel.

③ pLayer (PL): The i th bit is moved to bit position $P(i)$ by a constant permutation table P .

PRESENT can take keys of either 80 or 128 bits. Below is the key schedule of 80-bit version. The 80-bit key is stored in a register $K=k_{79}||k_{78}||\dots||k_0$. At round r ($1 \leq r \leq 31$), the 64-bit round key K^r is equal to the 64 leftmost bits of K . After K^r is extracted, K is rotated by 61-bit positions to the left, then a S-box is applied to the leftmost 4 bits of K and finally the round-counter r is Xored with bits $k_{19}||k_{18}||k_{17}||k_{16}||k_{15}$ of K .

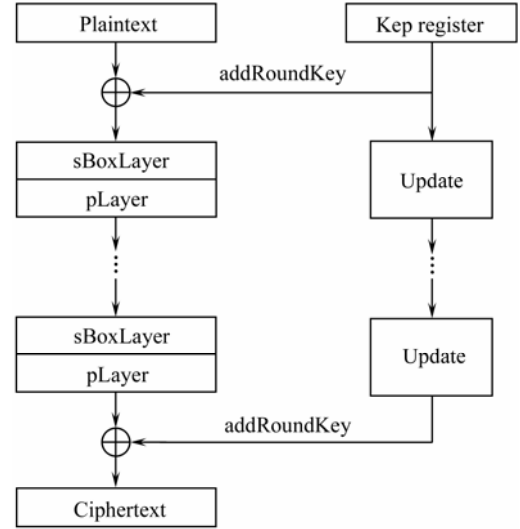


Fig. 3 Overview of PRESENT encryption algorithm

Table 1 introduces some notations used in this article. Bits are numbered from zero with bit zero on the right of a block.

Table 1 Notations used in FPP-FFA on PRESENT

Variable	Notations
Nibble	4-bit variables
r	round number
X, Y, A^i, B^i, C^i ($1 \leq i \leq 31$)	plaintext and ciphertext, the output of the i th round AK, SL, PL function
$X', Y', A^{i'}, B^{i'}, C^{i'}$	faulty variables of X, Y, A^i, B^i, C^i
$NX_j, NY_j, NA_j^i, NB_j^i, NC_j^i$ ($0 \leq j \leq 15$)	the j th nibble of X, Y, A^i, B^i, C^i
$X_j, Y_j, A_j^i, B_j^i, C_j^i$ ($0 \leq j \leq 63$)	the j th bit of X, Y, A^i, B^i, C^i
$\Delta X, \Delta Y, \Delta A^i, \Delta B^i, \Delta C^i$	fault difference of X, Y, A^i, B^i, C^i
$\Delta NX_j, \Delta NY_j, \Delta NA_j^i, \Delta NB_j^i, \Delta NC_j^i$ ($0 \leq j \leq 15$)	The j th nibble difference of X, Y, A^i, B^i, C^i
SX, SY, SA^i, SB^i, SC^i	faulty nibble index set of X, Y, A^i, B^i, C^i
Sx, Sy, Sa^i, Sb^i, Sc^i	faulty bit index set of X, Y, A^i, B^i, C^i

2.2 Fault Model

Below is the fault model of FPP-DFA on PRESENT.

① One random nibble fault is induced into the input of the r -2nd (29th) round. The attacker knows neither the location nor the concrete value of the fault.

② For any plaintext adaptively selected, two different ciphertexts generated by the same secret key are available, the right and the faulty ones. How to induce the specific fault is not the main concern of our article and many literatures on fault inductions are available in Ref. [8].

③ Only one master key is used in one attack.

Figure 4 shows the fault-propagation path of the above fault model with the maximal fault-propagation width, in which every faulty S-box output difference is (1111)₂.

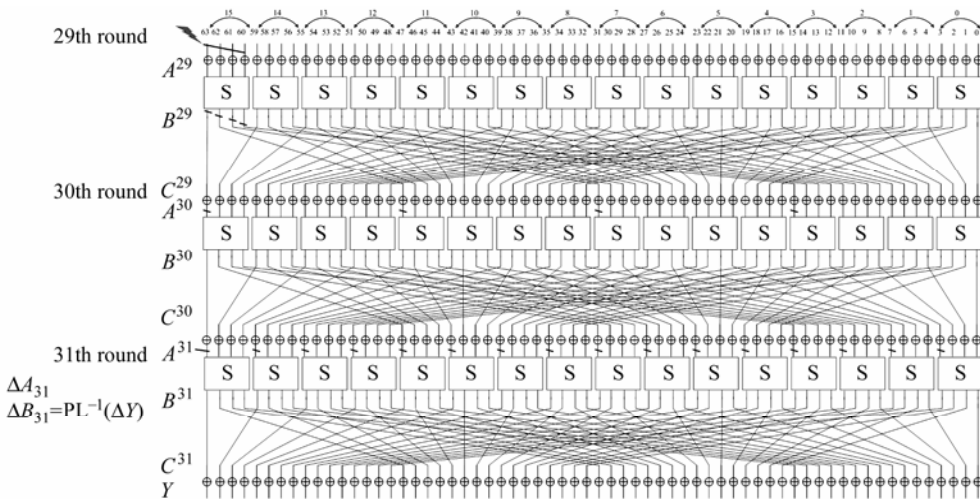


Fig. 4 One nibble fault-propagation path of PRESENT

2.3 FPP-DFA Procedure

The steps of FPP-DFA on PRESENT are listed below.

2.3.1 FPP learning

① Fault-propagation path learning

Based on the nibble fault model above, the fault-propagation path of the 16 possible fault locations (i) can be built. Table 2 shows a set of the faulty nibble and bit indexes for PRESENT with the maximal fault-propagation width after the 30th- and 31st-round substitution layers. From Table 2, at most 4 and 16 nibbles after the 30th- and 31st-round substitution are becoming faulty, and all the faulty nibble difference, $\Delta A^{31}=i/4+1$, so there are only four possible ΔA^{31} candidates for the 16 fault locations.

② Differential S-box pattern learning

According to Eq. (1), Table 3 displays all the possi-

ble value of f_2 when only one bit of f_1 is 1 and at least two bits of f_2 values 1. We can see that, after two rounds of fault propagation, at least 4 and at most 16 nibbles of the ciphertext are faulty, and each column value set of f_2 can be a distinguisher for f_1 .

2.3.2 Fault location and propagation path recovery

First, the 31st-round substitution output difference can be computed by equation $\Delta B^{31}=PL^{-1}(\Delta Y^{31})$. For each nonzero ΔNB_i^{31} , according to the four distinguishers of Table 3, the possible set of f_1 candidates, S_i can be deduced, and the intersection set of S_i is the final set of possible f_1 candidates. Then, according to the fault-propagation location distinguishers in the fourth column of Table 2, in which the set of possible fault locations can be obtained. Meanwhile, the more faulty nibbles in ΔB^{31} , the less candidates of ΔA^{31} can be deduced. After

Table 2 Fault-propagation path of PRESENT

i	$Sa^{30}(\Delta NA_i^{30})$	SA^{30}	$Sa^{31}(\Delta NA_i^{31})$
0	0,16,32,48(0001) ₂	0,4,8,12	0,16,32,48,4,20,36,52,8,24,40,56,12,28,44,60(0001) ₂
1	1,17,33,49(0010) ₂	0,4,8,12	0,16,32,48,4,20,36,52,8,24,40,56,12,28,44,60(0001) ₂
2	2,18,34,50(0100) ₂	0,4,8,12	0,16,32,48,4,20,36,52,8,24,40,56,12,28,44,60(0001) ₂
3	3,19,35,51(1000) ₂	0,4,8,12	0,16,32,48,4,20,36,52,8,24,40,56,12,28,44,60(0001) ₂
4	4,20,36,52(0001) ₂	1,5,9,13	1,17,33,49,5,21,37,53,9,25,41,57,13,29,45,61(0010) ₂
5	5,21,37,53(0010) ₂	1,5,9,13	1,17,33,49,5,21,37,53,9,25,41,57,13,29,45,61(0010) ₂
6	6,22,38,54(0100) ₂	1,5,9,13	1,17,33,49,5,21,37,53,9,25,41,57,13,29,45,61(0010) ₂
7	7,23,39,55(1000) ₂	1,5,9,13	1,17,33,49,5,21,37,53,9,25,41,57,13,29,45,61(0010) ₂
8	8,24,40,56(0001) ₂	2,6,10,14	2,18,34,50,6,22,38,54,10,26,42,58,14,30,46,62(0100) ₂
9	9,25,41,57(0010) ₂	2,6,10,14	2,18,34,50,6,22,38,54,10,26,42,58,14,30,46,62(0100) ₂
10	10,26,42,58(0100) ₂	2,6,10,14	2,18,34,50,6,22,38,54,10,26,42,58,14,30,46,62(0100) ₂
11	11,27,43,59(1000) ₂	2,6,10,14	2,18,34,50,6,22,38,54,10,26,42,58,14,30,46,62(0100) ₂
12	12,28,44,60(0001) ₂	3,7,11,15	3,19,35,51,7,23,39,55,11,27,43,59,15,31,47,63(1000) ₂
13	13,29,45,61(0010) ₂	3,7,11,15	3,19,35,51,7,23,39,55,11,27,43,59,15,31,47,63(1000) ₂
14	14,30,46,62(0100) ₂	3,7,11,15	3,19,35,51,7,23,39,55,11,27,43,59,15,31,47,63(1000) ₂
15	15,31,47,63(1000) ₂	3,7,11,15	3,19,35,51,7,23,39,55,11,27,43,59,15,31,47,63(1000) ₂

Table 3 Patterns of PRESENT differential S-box

$f_2(f_1=(0001)_2)$	$f_2(f_1=(0010)_2)$	$f_2(f_1=(0100)_2)$	$f_2(f_1=(1000)_2)$
	(0011) ₂ ,	(0101) ₂ ,	(0011) ₂ , (0111) ₂ ,
	(0101) ₂ ,	(0110) ₂ , (0111) ₂ ,	(1001) ₂ ,
(0011) ₂ , (0111) ₂ ,	(0110) ₂ ,	(1001) ₂ ,	(1011) ₂ , (1101) ₂ ,
(1001) ₂ , (1101) ₂	(1010) ₂ ,	(1010) ₂ ,	(1111) ₂
	(1100) ₂ , (1101) ₂ ,	(1100) ₂ , (1110) ₂	
	(1110) ₂		

the recovery of the possible fault locations, the fault-propagation path can be obtained easily with Table 2.

2.3.3 Fault-propagation path utilizing

① Extract the post-whitening key K^{32}

From above, limited candidates of ΔA^{31} can be computed. As ΔB^{31} can be computed by $\Delta B^{31} = PL^{-1}[Y] \oplus PL^{-1}[Y']$, then for each nonzero nibble of ΔNA_i^{31} , ΔNB_i^{31} , and NA_i^{31} satisfying

$$SL[NA_i^{31}] \oplus SL[NA_i^{31} \oplus \Delta NA_i^{31}] = \Delta NB_i^{31} \quad (2)$$

would be the correct NA_i^{31} . Then,

$$K_{P^{-1}(4*i+3)}^{32} \parallel K_{P^{-1}(4*i+2)}^{32} \parallel K_{P^{-1}(4*i+1)}^{32} \parallel K_{P^{-1}(4*i)}^{32} = SL[NA_i^{31}] \oplus Y_{P^{-1}(4*i+3)}^{32} \parallel Y_{P^{-1}(4*i+2)}^{32} \parallel Y_{P^{-1}(4*i+1)}^{32} \parallel Y_{P^{-1}(4*i)}^{32} \quad (3)$$

Applying Eq. (3), 4-bit of K^{32} can be recovered. As 4 to 16 nibbles of B^{31} are faulty, applying the technique above, 4 to 16 nibbles of K^{32} can be obtained, and full 64-bit of K^{32} can be recovered through more sample analysis.

② Extract the 31st-round key K^{31}

After K^{32} was deduced, ΔB^{30} can be computed by

$$\Delta B^{30} = SL^{-1}[PL^{-1}(Y)] \oplus SL^{-1}[PL^{-1}(Y')] \quad (4)$$

Using the distinguishers in Table 3, ΔA^{30} can be deduced. Then applying the same DFA technique above, 2 to 4 nibbles of K^{31} can be obtained. K^{31} can be recovered by analyzing more samples. Note that, as to the wrong K^{32} candidate, after the DFA analysis of about 4 faulty samples, some bits of K^{31} can always get an empty set, which can be used to eliminate the wrong K^{32} candidates.

③ Extract the master key K

Combining K^{31} , K^{32} , and the key schedule of PRESENT, the master key K can be recovered easily.

2.4 Complexity Analysis

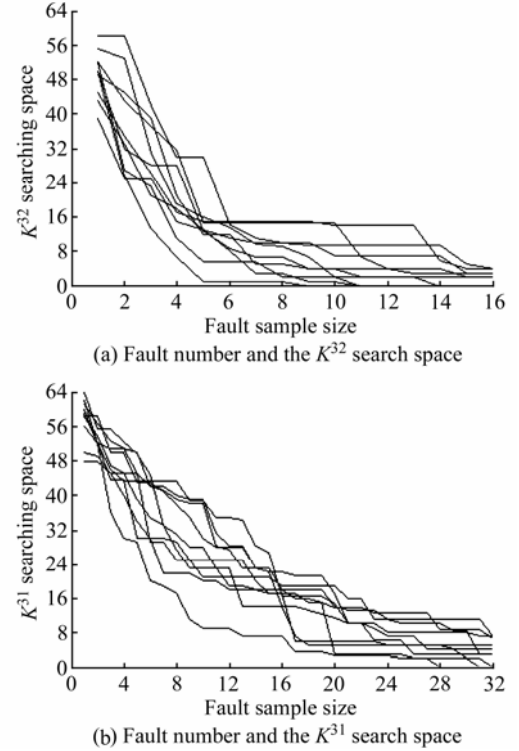
According to the patterns of the PRESENT differential S-box in Table 3, at least 2 to 4 nibbles of B^{30} become faulty, which can be used to deduce 2 to 4 nibbles of K^{31} , and at least 4 to 16 nibbles of B^{31} become faulty, which can be used to deduce 4 to 16 nibbles of K^{32} . As approximately two times analysis of the faulty nibble with the same index can recover one nibble key, about 8 faults can

obtain K^{32} and limited candidates of K^{31} , which are enough to recover the master key of PRESENT-80. Sixteen faults can obtain K^{32} and K^{31} , which are enough to obtain the master key of PRESENT-128.

2.5 Experiment Results

We have implemented our attack on a PC using Visual C++ 6.0 Compiler on a 1.81 GHz Athlon with 1 GB memory, and the fault induction was simulated by the computer software. The results are shown in Fig. 5.

In the PRESENT-80 attack, on average 8 samples can reduce the search space of K^{32} to $2^{7.6}$ (Fig. 5(a)). Combining the second step of the key extraction phase, wrong candidates of K^{32} can be eliminated, and the search space of the master key of PRESENT-80 can be further reduced to $2^{14.7}$. The results match with the previous theoretical analysis of Section 2.4 well.

**Fig. 5** Results of 10 times DFA on PRESENT

Next, we compared FPP-DFA with previous DFA with the encrypt procedure of PRESENT-80. Based on the fault model of one nibble fault injected between the r -2nd and the r -1st rounds, the work in Ref. [12] judged that the effective faulty samples are faulty, by whose 2 to 7 nibbles of B^{31} ; this judgment was performed using the S-box output differential distinguisher when only one bit of the S-box input nibble differential is 1. Then, the work in Ref. [12] deduced the nonzero nibble of ΔA^{31} one by one and combined ΔB^{31} to reduce the key search space of

K^{32} . Forty to 50 samples are needed to obtain K^{32} . However, this may discard faulty samples, of which 8 to 16 nibbles are faulty but still satisfied the one-bit distinguisher and the fault model.

FPP-DFA can make full use of the faulty samples. We started by learning the fault-propagation path of the model and utilized the one-bit distinguisher shown in Table 3 to deduce the possible four candidates of the 64-bit ΔA^{31} , finally recovered K^{32} by ΔA^{31} and ΔB^{31} . Moreover, after some candidates of K^{32} have been deduced, we use it to compute ΔB^{30} and apply the one-bit distinguisher of Table 3 again to predict ΔA^{30} . Then, the key search space of K^{31} can be reduced, and some wrong K^{32} candidates can also be eliminated.

Our technique can be adjusted to PRESENT-128 easily without changing the fault model. In FPP-DFA on PRESENT-128, it was shown that 16 samples could reduce the search space of K^{32} to $2^{1.8}$, the search space of K^{31} to $2^{16.3}$ (Fig. 5(b)) and the search space of the PRESENT-128 master key to $2^{21.1}$.

3 FPP-DFA on PRINTcipher

3.1 PRINTcipher Algorithm

PRINTcipher is a key-dependent SPN structure block cipher with b -bit blocks and b rounds, $b \in \{48, 96\}$, and the effective key length is of $5b/3$ bits. The $5b/3$ bits user-supplied key sk , which consisted of two subkey components $sk=sk_1||sk_2$, where sk_1 is a fixed b bits secret key and sk_2 is a $2b/3$ bits subkey to generate the key-dependent permutations and derive an additional of security via the secret algorithm variability. PRINTcipher has two variants determined by b : PRINTcipher-48 and PRINTcipher-96. Each encryption round consists of the following five steps.

① Key xor (KX): The b -bit current state of the cipher is Xored with a b -bit subkey sk_1 , and sk_1 is identical in all rounds.

② Linear diffusion (LD): Bit i of the current state is moved to bit position $P(i)$.

$$P(i) = \begin{cases} 3 \times i \bmod b-1 & \text{for } 0 \leq i \leq b-2 \\ b-1 & \text{for } i = b-1 \end{cases} \quad (5)$$

③ Round-counter RC_i xor (RX): The least significant n -bit ($n=\log_2 r$) of the cipher state is Xored with a round constant RC_i .

④ Keyed permutation (KP): The b -bit cipher state is divided into $b/3$ 3-bit nibbles. sk_2 is divided into $b/3$ sets of two bits, and every two-bit quantity $a_1||a_0$ is used

to pick one of four available permutations of the three input bits state nibble, especially the three input bits $c_2||c_1||c_0$, which are permuted by $a_1||a_0$

$$\text{input} : c_2 || c_1 || c_0 \Rightarrow \begin{cases} \text{case } a_1 || a_0 = 00 & \text{output} : c_2 || c_1 || c_0 \\ \text{case } a_1 || a_0 = 01 & \text{output} : c_1 || c_2 || c_0 \\ \text{case } a_1 || a_0 = 10 & \text{output} : c_2 || c_0 || c_1 \\ \text{case } a_1 || a_0 = 11 & \text{output} : c_0 || c_1 || c_2 \end{cases} \quad (6)$$

⑤ sBoxLayer (SL): $b/3$ identical 3-bit to 3-bit S-boxes are used in parallel.

Note that PRINTcipher has no key schedule, and the master key sk is used in every round.

Table 4 introduces some notations of FPP-DFA on PRINTcipher. The bits are also numbered from zero with bit zero on the right of a block.

Table 4 Notations used in FPP-DFA on PRINTcipher

Variable	Notations
Nibble	3-bit variables
r	round number of PRESENT
X, Y, A^i, B^i, C^i, D^i ($0 \leq i \leq b-1$)	the plaintext and ciphertext, the output of the i th round KX, LD, KP, SL function
$X', Y', A^{i'}, B^{i'}, C^{i'}, D^{i'}$	faulty variables of X, Y, A^i, B^i, C^i, D^i
$NX_j, NY_j, NA_j^i, NB_j^i, NC_j^i, ND_j^i$ ($0 \leq j \leq b-1/a$)	the j th nibble of X, Y, A^i, B^i, C^i, D^i
$X_j, Y_j, A_j^i, B_j^i, C_j^i, D_j^i$ ($0 \leq i, j \leq b-1$)	the j th bit of X, Y, A^i, B^i, C^i, D^i
$\Delta X, \Delta Y, \Delta A^i, \Delta B^i, \Delta C^i, \Delta D^i$	the fault difference of X, Y, A^i, B^i, C^i, D^i
$\Delta NX_j, \Delta NY_j, \Delta NA_j^i, \Delta NB_j^i, \Delta NC_j^i, \Delta ND_j^i$ ($0 \leq j \leq 15$)	the j th nibble difference of X, Y, A^i, B^i, C^i, D^i
$SX, SY, SA^i, SB^i, SC^i, SD^i$	faulty nibble index set of X, Y, A^i, B^i, C^i, D^i
$Sx, Sy, Sa^i, Sb^i, Sc^i, Sd^i$	faulty bit index set of X, Y, A^i, B^i, C^i, D^i

3.2 Fault Model

The fault model of FPP-DFA on PRINTcipher is the same as PRESENT, except that the injected faulty round m has subtle difference. As to PRINTcipher-48, m can be 46, and as to PRINTcipher-96, m can be 94 or 93. Next, we take FPP-DFA on PRINTcipher-48 as an example.

Figure 6 is the fault-propagation procedure of one nibble fault injecting into the 15th nibble before the 46th-round substitution with the maximized width, in which the S-box output difference is $(111)_2$.

3.3 FPP-DFA Procedure

The steps of FPP-DFA on PRINTcipher are listed below.

3.3.1 FPP learning

This phase is mainly about fault-propagation path learning.

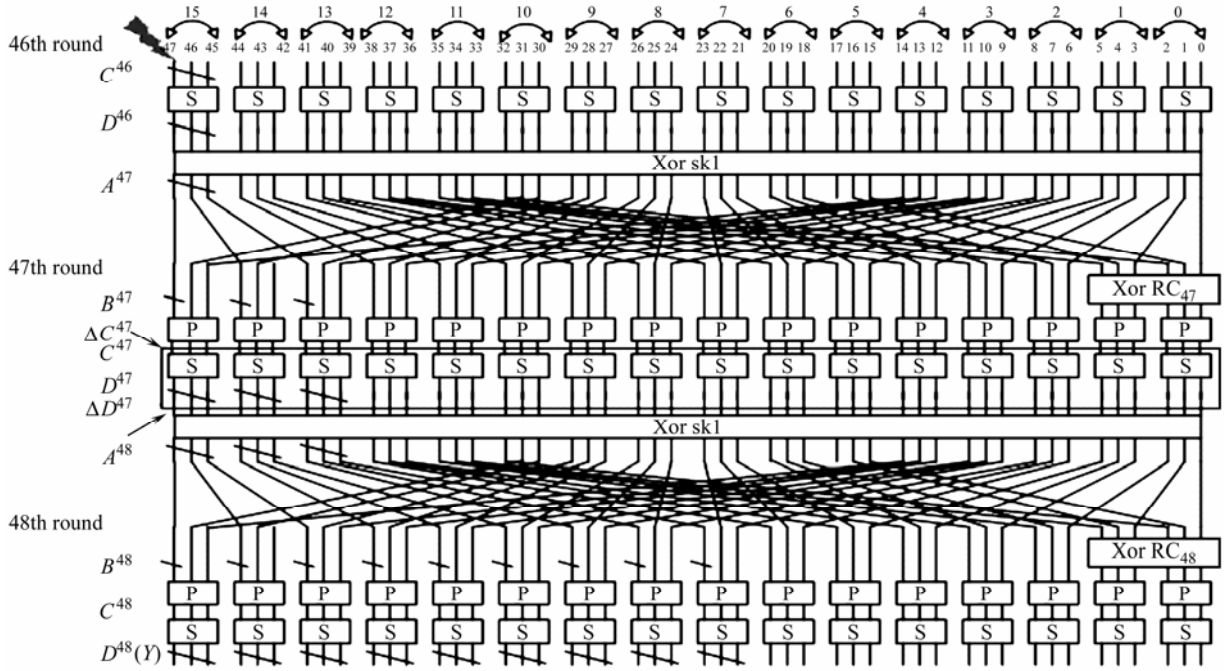


Fig. 6 One nibble fault-propagation path of PRINTcipher-48

Based on the fault model above, Table 5 shows the faulty nibble index set of PRINTcipher-48 with the maximal fault-propagation width after the 47th- and 48th-round substitutions and at most 3 and 9 nibbles after the 47th- and 48th-round substitution become faulty.

Table 5 Fault-propagation path of PRINTcipher-48

i	SD^{47}	SD^{48}	i	SD^{47}	SD^{48}
0	0,1,2	0,1,2,3,4,5,6,7,8	8	8,9,10	8,9,10;11,12,13;14,15,0
1	3,4,5	9,10,11;12,13,14;15,0,1	9	11,12,13	1,2,3;4,5,6;7,8,9
2	6,7,8	2,3,4;5,6,7,8,9,10	10	14,15,0	10,11,12;13,14,15;0,1,2
3	9,10,11	11,12,13;14,15,0;1,2,3	11	1,2,3	3,4,5;6,7,8,9,10,11
4	12,13,14	4,5,6;7,8,9;10,11,12	12	4,5,6	12,13,14;15,0,1;2,3,4
5	15,0,1	13,14,15;0,1,2,3,4,5	13	7,8,9	5,6,7,8,9,10;11,12,13
6	2,3,4	6,7,8,9,10,11;12,13,14	14	10,11,12	14,15,0;1,2,3;4,5,6
7	5,6,7	15,0,1;2,3,4;5,6,7	15	13,14,15	7,8,9;10,11,12;13,14,15

3.3.2 Fault location and propagation path recovery

Take injecting fault into the 15th nibble of the 46th-round substitution input as an example. According to the ciphertext difference, a distinct faulty nibble index set SD^{48} can be computed. Suppose $SD^{48} = \{7, \dots, 15\}$, according to Table 5, we can deduce that the faulty nibble index is 15.

Then, it is stored as an effective fault sample. Interestingly, if the exact injected fault location i of the 46th round can be deduced, either all three nibbles of ΔD^{47} with consecutive indexes ($SD^{47} = \{13, 14, 15\}$) become faulty or two nibbles of ΔD^{47} with nonconsecutive indexes ($SD^{47} = \{13, 15\}$) become faulty, which also means

that ΔND_i^{46} can be either $(101)_2$ or $(111)_2$. After the fault location is deduced, both ΔB^{47} and ΔB^{48} can be obtained.

3.3.3 Fault-propagation path utilizing

① Extract the permutation related key sk_2

After the fault location and fault-propagation path is deduced, sk_2 can be obtained by the following algorithm.

$$\Delta C^{48} \leftarrow SL^{-1}[D^{48}] \oplus SL^{-1}[D^{48'}]$$

$$\Delta B^{48} \text{ can be deduced by } \Delta D^{48}$$

For each index i in SD^{48}

```

{
   $j \leftarrow$  faulty index bit of  $\Delta NB_i^{48}$ 
  If ( $j == 2$ )
  {
     $\Delta NB_i^{48} \leftarrow (100)_2$ 
    If ( $\Delta NB_i^{48} == (100)_2$ )
     $sk_{2, 2*i} = (0)_2$ 
    else if ( $\Delta NB_i^{48} == (010)_2$ )
     $sk_{2, 2*i+1} || sk_{2, 2*i} = (01)_2$ 
    else if ( $\Delta NB_i^{48} == (000)_2$ )
     $sk_{2, 2*i+1} || sk_{2, 2*i} = (11)_2$ 
  }
  If ( $j == 1$ )
  {
     $\Delta NB_i^{48} \leftarrow (010)_2$ 
    If ( $\Delta NB_i^{48} == (010)_2$ )
     $sk_{2, 2*i+1} || sk_{2, 2*i} = \{(00)_2, (11)_2\}$ 
    else if ( $\Delta NB_i^{48} == (100)_2$ )
     $sk_{2, 2*i+1} || sk_{2, 2*i} = \{01\}_2$ 
    else if ( $\Delta NB_i^{48} == (001)_2$ )

```

```

    }
    If(j==0)
    {
         $\Delta \text{NB}_i^{48} \leftarrow (001)_2$ 
        If ( $\Delta \text{NB}_i^{48} == (001)_2$ )
             $\text{sk}_2, 2^{*i+1} = \{0\}_2$ 
        else if ( $\Delta \text{NB}_i^{48} == (100)_2$ )
             $\text{sk}_2, 2^{*i+1} \parallel \text{sk}_2, 2^{*i} = \{11\}_2$ 
        else if ( $\Delta \text{NB}_i^{48} == (010)_2$ )
             $\text{sk}_2, 2^{*i+1} \parallel \text{sk}_2, 2^{*i} = \{10\}_2$ 
    }
}

```

Applying the above algorithm, after one-sample analysis, at most 18 bits of sk_2 can be recovered, and full 32 bits of sk_2 can be obtained with more sample analysis.

② Extract the secret key sk_1

Once sk_2 was recovered, ΔD^{47} can be computed by $\text{SL}[\text{NC}_{15}^{47}] \oplus \text{SL}[\text{NC}_{15}^{47} \oplus \Delta \text{NC}_{15}^{47}] = \Delta \text{ND}_{15}^{47}$ (7)

Take recovering $\text{sk}_{1,47} \parallel \text{sk}_{1,46} \parallel \text{sk}_{1,45}$ as an example, according to Fig. 6, ΔNB^{47} can be computed, and $\Delta \text{NB}_{15}^{47} = (100)_2$, then $\Delta \text{NC}_{15}^{47}$ can be deduced by $\text{KP}(\Delta \text{NB}_{15}^{47})$. Candidates of NC_{15}^{47} satisfying

$$\text{SL}[\text{NC}_{15}^{47}] \oplus \text{SL}[\text{NC}_{15}^{47} \oplus \Delta \text{NC}_{15}^{47}] = \Delta \text{ND}_{15}^{47} \quad (8)$$

would be the possible one. Then,

$$\text{sk}_{1,47} \parallel \text{sk}_{1,46} \parallel \text{sk}_{1,45} = \text{SL}[\text{NC}_{15}^{47}] \oplus \text{LD}^{-1}(\text{RX}^{-1}(\text{KP}^{-1}(\text{SL}^{-1}[D^{48}]))) \quad (9)$$

Applying the same method above, after one-sample analysis, at most 9 bits of sk_1 can be recovered, and after more sample analysis, all the 48 bits of sk_1 can be obtained.

③ Extract the master key sk

The master key sk equals to $\text{sk}_1 \parallel \text{sk}_2$.

3.4 Complexity Analysis

① Probability of effective fault location deducing

Suppose the S-box index is a , the input and output difference are f_1 and f_2 , and a, f_1 , and f_2 are all 3-bit variables that satisfy Eq. (1).

Table 6 displays all the possible candidates of f_2 when only one bit of f_1 is 1. According to Section 3.3, if the fault location i was deduced, ΔND_i^{46} can be either $(101)_2$ or $(111)_2$, and the probability is 2/7. The first bit of the output difference ΔND_j^{47} propagated by the first bit of ΔND_i^{46} should be 1, and its probability is 16/24. The third output difference ΔND_j^{47} propagated by the third bit of ΔND_i^{46} should be 1, and its probability is 16/24. Thus, the overall fault location deducing probability is 12.7%.

② Complexity analysis of sk_2 deducing

As on average 5.5 nibbles of B^{48} have become faulty, according to Section 3.3, if one nibble faulty difference

Table 6 Patterns of PRINTcipher differential S-box

a	$f_2(f_1=(001)_2)$	$f_2(f_1=(010)_2)$	$f_2(f_1=(100)_2)$
$(000)_2$	$(001)_2$	$(011)_2$	$(111)_2$
$(001)_2$	$(001)_2$	$(111)_2$	$(101)_2$
$(010)_2$	$(101)_2$	$(011)_2$	$(110)_2$
$(011)_2$	$(101)_2$	$(111)_2$	$(100)_2$
$(100)_2$	$(011)_2$	$(010)_2$	$(111)_2$
$(101)_2$	$(011)_2$	$(110)_2$	$(101)_2$
$(110)_2$	$(111)_2$	$(010)_2$	$(110)_2$
$(111)_2$	$(111)_2$	$(110)_2$	$(100)_2$

after the KP layer changed, 2-bit of sk_2 can be deduced directly. Otherwise, either one bit of sk_2 can be deduced directly or two bits of sk_2 are equal. Thus, on average 5/3 bits of sk_2 can be deduced for each faulty B^{48} nibble. Then, about 3.5 faulty samples are needed to recover sk_2 .

③ Complexity analysis of sk_1 deducing

Once the fault location is known, on average 2.5 nibbles of ΔNB^{47} become faulty and can be used to deduce sk_1 . From Table 6, if only one bit of ΔNB^{47} is 1, on average 2 candidates of a can be deduced, the key search space of one sk_1 nibble can be reduced from 8 to 2, and two times analysis of the nibble with the same index can recover the sk_1 nibble. 1.25 nibbles of sk_1 can be deduced and 12.8 samples are required to obtain sk_1 .

3.5 Experiment Results

In Fig. 7(a), there is the effective fault number of 10 times FPP-DFA experiment on PRINTcipher-48 with 10 000 samples. The average fault location deducing probability is 13.1%, which is close to the theoretical value of 12.7%.

Figure 7(b)-(d) shows the relationships between the fault number and the key search space of sk_2 , sk_1 , and sk . Four effective faults can reduce the sk_2 key search space from 2^{32} to $2^{9.1}$, 12 effective faults can reduce the sk_1 search space from 2^{48} to $2^{2.1}$, 12 effective faults can reduce the master key search space from 2^{80} to $2^{13.7}$, and the results well agree with previous theoretical analysis.

Under the r -2nd round fault model, we have applied the FPP-DFA on PRINTcipher-96. On average 24 effective faults can reduce the search space of the 160-bit master key to $2^{22.8}$. Meanwhile, based on the fault model of injecting one nibble into the r -3rd round substitution, we have also applied the FPP-DFA on PRINTcipher-96. On average 8 effective faults can reduce the search space of the 160-bit master key to $2^{18.7}$.

4 Conclusion

We have proposed a novel FPP-DFA techniques on SPN structure block ciphers with bitwise permutation, ap-

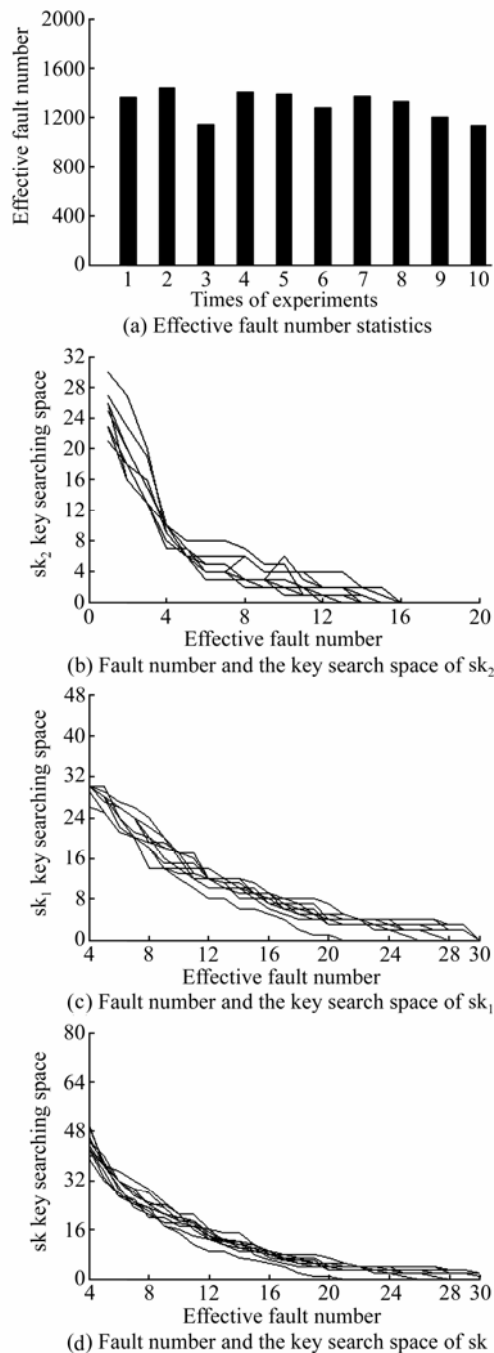


Fig. 7 Experimental results of PRINTcipher-48 attack

plied it to PRESENT and PRINTcipher, and proven its efficiency through concrete simulated experiments. We conclude that implementing SPN structure block ciphers with bitwise permutation should take certain countermeasures against fault-based attacks; at least the last three rounds of PRESENT-80/128, PRINTcipher-48, and at least the last four rounds of PRINTcipher-96 should be protected.

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